

SECTION: A

1) $b > 1$ ✓

2) $a > 0$ ✓

3) $a > 0$ ✓

4) $b > 9^2$ ✓

5) $d > \text{May or may not be consistent}$
 If $|A| = 0$ and $(\text{adj } A)B = 0$.

6) $d > 5$. ✓

7) $b > 24$. ✓

8) $d > [0, 1]$ ✓

9) $d > -2 \coth\left(\frac{y}{2}\right)$

10) $b > 2$ ✓

11) $b > \text{at only two points.}$

12) $c > \text{local max value is } -2$

13) $d > x \in (2, \infty) \cup (-\infty, 0)$

14) $b > \frac{ax^2}{2} + bx$

15) $c > \log |\sin x + \sqrt{\sin^2 x + 1}| + c$

16) $c > \sin^{-1}(e^x)$

17) $b > x + y \geq 5, x + 3y \leq 9.$

18) $c > (0, 8)$

19) $c > A \text{ is true, } R \text{ is false.}$

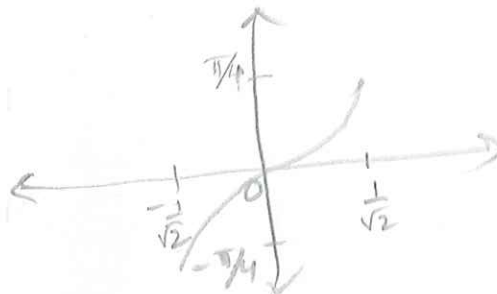
20) $c > A \text{ is true, } R \text{ is true.}$

SECTION - B

$$\begin{aligned}
 21) a) \tan^{-1} \sin\left(-\frac{\pi}{2}\right) + \tan^{-1}\left(\tan\frac{\pi}{3}\right) + \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) \\
 = \tan^{-1}(-1) + \tan^{-1}\left\{\tan\left[\pi - \frac{\pi}{3}\right]\right\} + \left(-\frac{\pi}{6}\right) + \frac{\pi}{3} \\
 = -\frac{\pi}{4} + \left(-\frac{\sqrt{3}}{3}\right) - \frac{\pi}{6} + \frac{\pi}{3} = -\frac{\pi}{4} - \frac{\pi}{6} \\
 = -\frac{5\pi}{12}
 \end{aligned}$$

(OR)

21) b) Range of $\csc^{-1}x \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
 Range of $\cot^{-1}x \rightarrow (0, \pi)$



$$22) a) U = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$$

$$V = \sin^{-1}(2x\sqrt{1-x^2})$$

Put $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

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$$U = \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)$$

$$= \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta})$$

$$= \tan^{-1}(\tan\theta)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$\theta = \theta = \sin^{-1}x$$

$$= \sin^{-1}(\sin 2\theta)$$

D w.r.t x

$$= 2\theta$$

$$\frac{dU}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$V = a(\sin^{-1}x)$$

D w.r.t x

$$\frac{dV}{dx} = a\left(\frac{1}{\sqrt{1-x^2}}\right)$$

$$\therefore \frac{dU}{dV} = \frac{dU/dx}{dV/dx} = \frac{\frac{1}{\sqrt{1-x^2}}}{\frac{a}{\sqrt{1-x^2}}} = \frac{1}{a}$$

(OR)

$$22) b) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Putting $x=5$

$$f'(5) = \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$$

using $f(x+y) = f(x) \cdot f(y)$

$$\begin{aligned}
 f'(5) &= \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(5) \cdot f(h) - f(5)}{h} \\
 &= \lim_{h \rightarrow 0} f(5) \left\{ \frac{f(h) - 1}{h} \right\}
 \end{aligned}$$

Put $f(5) = 2$

$$= \lim_{h \rightarrow 0} 2 \times \left\{ \frac{f(h) - 1}{h} \right\} \text{ --- ①}$$

We have $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

Put $x = 0$

$$= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$f(0+h) = f(h)$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0) f(h) - f(0)}{h}$$

$$\therefore f(x+y) = f(x) f(y)$$

put $f'(0) = 3$

$$3 = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \text{ --- ②}$$

Given $f(x+y) = f(x) f(y)$

and $f(5) = 2$

Put $x = 0, y = 5$

$$f(0+5) = f(0) f(5)$$

$$f(5) = f(0) f(5)$$

$$2 = f(0) \cdot 2$$

$$f(0) = 1$$

$\therefore$ Eqn ② $\Rightarrow$

$$3 = \lim_{h \rightarrow 0} \frac{f(h) - 1}{h}$$

$\therefore$ Eqn ① $\Rightarrow$

$$f'(5) = 2 \cdot \lim_{h \rightarrow 0} \left\{ \frac{f(h) - 1}{h} \right\}$$

$$= 2 \times 3$$

$$= 6 //$$

$$23) f(1) = \frac{1+1}{2} = \frac{2}{2} = 1 \quad (\text{since } 1 \text{ is odd, } f(n) = \frac{n+1}{2})$$

$$f(2) = \frac{2}{2} = 1 \quad (\text{since } 2 \text{ is even, } f(n) = \frac{n}{2})$$

$$\therefore f(1) = 1 = f(2)$$

but $1 \neq 2$, It is NOT one-one. //

* Surjective: (ONTO)

Let $f(n) = y$, such that $y \in \mathbb{N}$

When n is odd

$$y = \frac{n+1}{2}$$

$$2y = n+1$$

$$n = 2y - 1$$

Hence $y \in \mathbb{N}$,
 $n \in \mathbb{N}$

When n is even

$$y = \frac{n}{2}$$

$$2y = n$$

$$n = 2y$$

$$y \in \mathbb{N}$$

$$n \in \mathbb{N}$$

Thus for every $y \in \mathbb{N} \quad \exists n \in \mathbb{N}$ such that $f(n) = y$
 f is onto //

$\therefore f$ is NOT bijective //

$$24) \int \frac{1}{x} \sqrt{\frac{x+a}{x-a}} dx$$

$$= \int \frac{1}{x} \sqrt{\frac{x+a}{x-a}} \times \frac{x+a}{x+a} dx$$

$$= \int \frac{1}{x} \sqrt{\frac{(x+a)^2}{x^2 - a^2}} dx$$

$$= \int \frac{1}{x} \cdot \frac{(x+a)}{\sqrt{x^2 - a^2}} dx$$

$$= \int \frac{1}{x} \cdot \left[\frac{x}{\sqrt{x^2 - a^2}} \right] dx + \int \frac{a}{x \sqrt{x^2 - a^2}} dx$$

$$= \int \frac{1}{\sqrt{x^2 - a^2}} dx + a \int \frac{1}{x \sqrt{x^2 - a^2}} dx$$

$$= \log|x + \sqrt{x^2 - a^2}| + \frac{a}{a} \sec^{-1}\left(\frac{x}{a}\right) + C$$

$$= \log|x + \sqrt{x^2 - a^2}| + \sec^{-1}\left(\frac{x}{a}\right) + C //$$

25) $f(x) = \begin{cases} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} & : \text{if } x \neq 0 \\ e^{\frac{1}{x}} + 1 & : \text{if } x = 0 \end{cases}$

$$\begin{aligned} e^0 &= 1 \\ e^{+\infty} &= \infty \\ e^{-\infty} &= 0 \end{aligned}$$

LHL = $\lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} = \lim_{h \rightarrow 0} \frac{e^{-\frac{1}{h}} - 1}{e^{-\frac{1}{h}} + 1} = \frac{0 - 1}{0 + 1} = -1$

RHL = $\lim_{x \rightarrow 0^+} \left\{ \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} \right\} = \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1} = \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} [1 - e^{-\frac{1}{h}}]}{e^{\frac{1}{h}} [1 + e^{-\frac{1}{h}}]}$

LHL $\neq$ RHL

$\therefore f(x)$ is discontinuous at $x = 0$

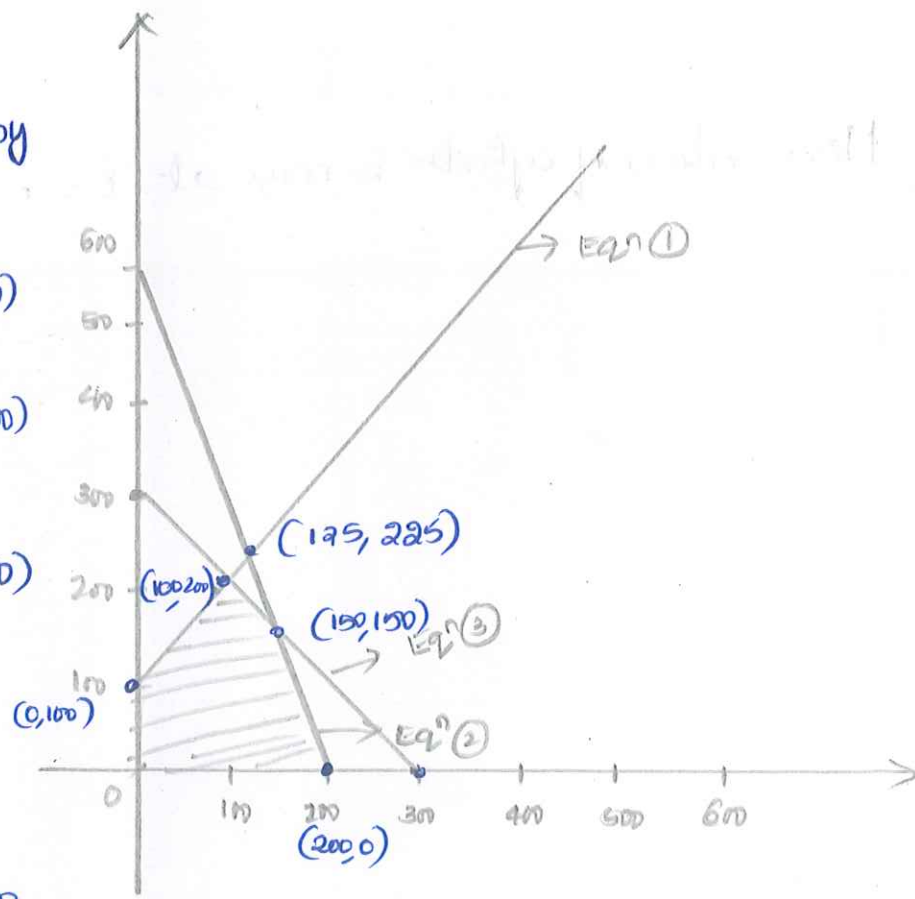
$= \lim_{h \rightarrow 0} \frac{1 - e^{-\frac{1}{h}}}{1 + e^{-\frac{1}{h}}} = \frac{1 - 0}{1 + 0} = 1$

SECTION : C

26) $Z = 200x + 120y$

$x - y \geq -100$ — (1) ; $x - y = -100$ $(0, 100), (900, 0)$
 $3x + y \leq 600$ — (2) ; $3x + y = 600$ $(0, 600), (200, 0)$
 $x + y \leq 300$ — (3) ; $x + y = 300$ $(0, 300), (300, 0)$

Corner Point (x, y)	$Z = 200x + 120y$
(0, 0)	0
(0, 100)	$200(0) + 120(100) = 12,000$
(100, 200)	$200(100) + 120(200) = 44,000$
(150, 150)	$200(150) + 120(150) = 48,000$
(200, 0)	$200(200) + 120(0) = 40,000$



$Z_{max} = 48,000$ at $x = 150$
 $y = 150$

27)



Let $L \rightarrow$ length $V \rightarrow$ volume of cylinder
 $w \rightarrow$ width
 $P \rightarrow$ perimeter

$$P = 2(L+w) = 36$$

$$h = w, r = L$$

$$L + w = 18$$

$$V = \pi r^2 h = \pi r^2 (w) = \pi r^2 (18 - L) = \pi r^2 (18 - r)$$

$$V = 18\pi r^2 - \pi r^3$$

D wrt r

$$\frac{dV}{dr} = 36\pi r - 3\pi r^2$$

$$\text{for max volume } \frac{dV}{dr} = 0 \Rightarrow 36\cancel{r} = 3\cancel{r} r$$

$$12 = r$$

$$r = 0, 12 \quad (L=r)$$

$$\frac{d^2V}{dr^2} = 36\pi - 6\pi r$$

$$\text{at } r = 12$$

$$= 36\pi - 72\pi$$

$$= -36\pi < 0$$

Here volume of cylinder is max at $L=r=12$ cm and $w=6$ cm //

$$28) f(x) = -3 \log(1+x) + 4 \log(2+x) - \frac{4}{2+x}$$

$$f'(x) = -\frac{3}{1+x} + \frac{4}{2+x} - \frac{4}{(2+x)^2} = \frac{x(x+4)}{(1+x)(2+x)^2}$$

$$f'(x) = 0 \Rightarrow x = 0 \text{ or } x = -4$$

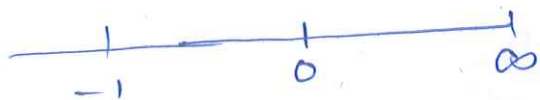
$$x = -4 \notin (-1, \infty)$$

$x = 0$ critical point

$$\text{Dom } f(x) = (-1, \infty)$$

$$0 \uparrow (-4, -2) \cup (-2, -1) \cup (0, \infty)$$

$$0 \downarrow (-\infty, -4) \cup (-1, 0)$$

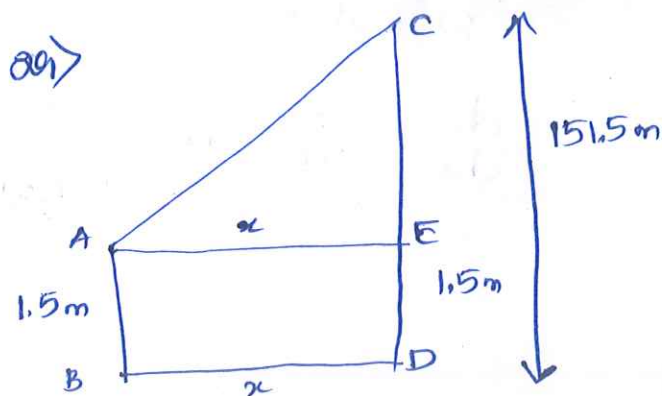


i) $(-1, 0)$; $x+4, (1+x), (2+x)^2 > 0$ (+)
 All are positive, while x is negative

$\therefore f'(x) < 0$. strictly decreasing.

ii) $(0, \infty)$; for any $x > 0$, $f'(x)$ are strictly positive

$\therefore f'(x) > 0$. strictly increasing. //



CD $\rightarrow$ height of the kite
 AB $\rightarrow$ height of the boy

$$\text{Given } \frac{dx}{dt} = 10 \text{ m/s}$$

$$EC = 150 \text{ m}$$

$$AC = 250 \text{ m}$$

$$AE^2 = EC^2 + CE^2$$

$$y^2 = (150)^2 + x^2$$

D wof x

$$2y \frac{dy}{dt} = 0 + 2x \frac{dx}{dt}$$

$$\frac{dy}{dt} = \frac{x}{y} \left(\frac{dx}{dt} \right) \quad \text{--- (1)}$$

$$\frac{dy}{dt} = \frac{200}{250} (10) = \frac{200}{25} = 8 \text{ m/s}$$

=

$$\begin{aligned} \text{When } y &= 250 \text{ m, } x^2 + 150^2 = y^2 \\ x^2 + (150)^2 &= (250)^2 \\ x^2 &= 62500 - 22500 \\ x^2 &= 40000 \\ x &= 200 \text{ m} \end{aligned}$$

$$30) a) y\sqrt{1-x^2} + x\sqrt{1-y^2} = 1$$

$$\text{Put } x = \sin\theta, y = \sin\phi$$

$$\theta = \sin^{-1}x$$

$$\sin\phi (\cos\theta) + \sin\theta \cos\phi = 1$$

$$\phi = \sin^{-1}y$$

$$\sin(\theta + \phi) = 1$$

$$\theta + \phi = \sin^{-1}1$$

$$\sin^{-1}x + \sin^{-1}y = \frac{\pi}{2}$$

To wot x

$$\frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-y^2}} \left(\frac{dy}{dx} \right) = 0$$

$$\frac{1}{\sqrt{1-y^2}} \frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

(OR)

$$\therefore \frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} //$$

$$30) b) y = x^3 \cdot (\sin x)^x + \sin^{-1}\sqrt{3x}$$

$$y = U + V$$

To wot x

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} \quad \text{--- (1)}$$

$$\text{let } U = x^3 \cdot (\sin x)^x$$

Take log on B.S

$$\log u = \log (x^3 \cdot (\sin x)^x)$$

$$= \log x^3 + \log (\sin x)^x$$

$$= 3 \log x + x \cdot \log (\sin x)$$

To wot x

$$\frac{1}{u} \frac{du}{dx} = \frac{3}{x} + x \left\{ \frac{1}{\sin x} \cos x \right\} + \log (\sin x) \cdot 1$$

$$\frac{du}{dx} = u \left\{ \frac{3}{x} + x \cot x + \log (\sin x) \right\} \quad \text{--- (2)}$$

Substitute (2), (3) in (1)

//

let

$$v = \sin^{-1}\sqrt{3x}$$

To wot x

$$\frac{dv}{dx} = \frac{1}{\sqrt{1-(\sqrt{3x})^2}} \cdot \left\{ \frac{3}{2\sqrt{3x}} \right\} \quad \text{--- (3)}$$

$$31) a) \int e^{\cot^{-1}x} \left\{ \frac{1-x+x^2}{1+x^2} \right\} dx$$

$$\text{let } I = \int e^{\cot^{-1}x} \left\{ \frac{(1+x^2) - x}{1+x^2} \right\} dx = \int e^{\cot^{-1}x} \left[\frac{1+x^2}{1+x^2} + \frac{-x}{1+x^2} \right] dx$$

$$I = \int e^{\cot^{-1}x} \left\{ 1 - \frac{x}{1+x^2} \right\} dx$$

$$\cot^{-1}x = t$$

$$x = \cot t$$

$$\text{D w.r.t } x$$

$$= \int e^t \left\{ 1 - \frac{\cot t}{1+\cot^2 t} \right\} (c \operatorname{osec}^2 t) dt \quad 1 = -\operatorname{cosec}^2 t \cdot \frac{dt}{dx}$$

$$dx = -\operatorname{cosec}^2 t dt$$

$$= \int e^t \left\{ \cancel{\operatorname{cosec}^2 t} + \frac{\cot t \cdot \cancel{\operatorname{cosec}^2 t}}{1+\cot^2 t} \right\} (-dt)$$

$$= \int e^t \{ \operatorname{cosec}^2 t + \cot t \} (-dt)$$

$$= \int e^t \{ -\operatorname{cosec}^2 t + \cot t \} dt$$

$$= e^t \cot t + c$$

$$= e^{\cot^{-1}x} \cot(\cot^{-1}x) + c$$

$$= e^{\cot^{-1}x} \cdot x + c //$$

$$\text{WKT } \int e^x [f(x) + f'(x)] dx$$

$$= e^x f(x)$$

$$(OR) 31) b) \int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})} dx //$$

$$I = \int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})} dx$$

$$= \int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{e^{2x} - e^{-2x}} dx$$

$$= \int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{e^{2x}}{e^{4x} - 1} dx$$

$$(e^x + e^{-x})(e^x - e^{-x}) = e^{2x} - e^{-2x}$$

$$(a+b)(a-b) = a^2 - b^2$$

$$\text{Let } e^{2x} = t$$

$$\text{D w.r.t } x$$

$$e^{2x} \cdot 2 = \frac{dt}{dx}$$

$$e^{2x} dx = \frac{1}{2} dt$$

when $x = \log \sqrt{2}$; $t = e^{2x} = e^{2 \log \sqrt{2}} = e^{\log(\sqrt{2})^2} = 2$
 11/4 $x = \log \sqrt{3}$; $t = 3$,

$$\begin{aligned} I &= \int_2^3 \frac{1}{(t^2-1)} \frac{dt}{2} = \frac{1}{2} \int_2^3 \frac{1}{t^2-1} dt \\ &= \frac{1}{2} \left\{ \frac{1}{2} \log \left| \frac{t-1}{t+1} \right| \right\}_2^3 \\ &= \frac{1}{4} \log \left| \frac{t-1}{t+1} \right|_2^3 \\ &= \frac{1}{4} \log \left(\frac{2}{4} \right) - \frac{1}{4} \log \left(\frac{1}{3} \right) \\ &= \frac{1}{4} \log \left(\frac{3}{2} \right) // \end{aligned}$$

SECTION-D

32) $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$

$$\begin{aligned} |A| &= 1 \begin{vmatrix} -1 & -2 \\ -1 & 1 \end{vmatrix} - 2 \begin{vmatrix} -2 & -2 \\ 0 & 1 \end{vmatrix} + 0 \begin{vmatrix} -2 & -1 \\ 0 & -1 \end{vmatrix} \\ &= 1 \neq 0, \quad \boxed{A^{-1} \text{ exists}} \end{aligned}$$

$$\text{adj}(A) = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ -4 & 2 & 3 \end{bmatrix}^T = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj} A = \frac{1}{1} \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\left. \begin{aligned} x - 2y &= 10 \\ 2x - y - z &= 8 \\ 3x - 2y + 1z &= 7 \end{aligned} \right\} \text{eqn form}$$

$$X = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}^T \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 & 2 & 2 \\ -2 & 1 & 1 \\ 4 & 2 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ -3 \end{bmatrix}$$

$$x = 0, y = -5, z = -3 //$$

$$\begin{bmatrix} 1 & -2 & 0 \\ 2 & -1 & -1 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \\ 7 \end{bmatrix}$$

$$(A^T) X = B$$

$$X = (A^T)^{-1} \cdot B$$

$$X = (A^{-1})^T \cdot B$$

33) a) $R = \{(x, y) : |x - y| \text{ is div by a prime num 'p', } x, y \in \mathbb{Z}\}$

1) let $x \in \mathbb{Z}$, $|x - x| = 0$ divisible by 'p'

$\therefore (x, x) \in R$ for all $x \in \mathbb{Z}$

$\therefore R$ is reflexive.

ii) 2) If $(x, y) \in R \Rightarrow |x - y|$ is div by p

$\Rightarrow |-(y - x)|$ is div by p

$\Rightarrow |y - x|$ is also div by p

$\Rightarrow (y, x) \in R$. Thus R is symmetric.

iii) 3) If $(x, y) \in R$ and $(y, z) \in R$

$|x - y|$ is div by p $\Rightarrow x - y = \pm k_1 p$ — (1) $k_1 \in \mathbb{Z}$

$|y - z|$ is div by p $\Rightarrow y - z = \pm k_2 p$ — (2) $k_2 \in \mathbb{Z}$

Add (1) & (2)

$$(x - y) + (y - z) = \pm k_1 p \pm k_2 p$$

$$(x - z) = p(\pm k_1 \pm k_2)$$

$$x - z = p(k_3)$$

$|x - z|$ is div by 'p'.

$\therefore R$ is transitive.

R is reflexive, sym, transitive

R is an equivalence relation //

b) $f: \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = x^2 + x + 1$

* One-One: let $x_1, x_2 \in \mathbb{R}$

$$\text{If } f(x_1) = f(x_2)$$

$$x_1^2 + x_1 + 1 = x_2^2 + x_2 + 1$$

$$x_1^2 + x_1 + 1 - x_2^2 - x_2 - 1 = 0$$

$$(x_1^2 - x_2^2) + (x_1 - x_2) + 1 - 1 = 0$$

$$(x_1 - x_2)(x_1 + x_2) + (x_1 - x_2) + 0 = 0$$

$$(x_1 - x_2) \{ (x_1 + x_2) + 1 \} = 0$$

$$x_1 = x_2 \text{ (or) } x_1 + x_2 + 1 = 0$$

$$\boxed{x_1 = -1(x_2 + 1)}$$

$\therefore$ It is NOT one-one.

* ONTO:

$$\text{let } f(x) = y$$

$$x^2 + x + 1 = y$$

$$x^2 + x + 1 - y = 0$$

$$x = \frac{-1 \pm \sqrt{1 - 4(1)(1-y)}}{2}$$

$$= \frac{-1 \pm \sqrt{1 - 4 + 4y}}{2}$$

$$= \frac{-1 \pm \sqrt{-3 + 4y}}{2}$$

Clearly

$$-3 + 4y \geq 0$$

$$4y \geq 3$$

$$y \geq 3/4$$

$$\text{Range} = \left[\frac{3}{4}, \infty \right)$$

$$\text{Codomain} = \mathbb{R}$$

$$\therefore \text{Range} \neq \text{Cod}$$

$f(x)$ is NOT onto

$\therefore f(x)$ is NOT bijective

Now; $f(x) = 3$

$$x^2 + x + 1 = 3$$

$$x^2 + x - 2 = 0$$

$$x^2 + 2x - x - 2 = 0$$

$$x(x+2) - 1(x+2) = 0$$

$$x = 1, x = -2 //$$

$$34) a) f(x) = |x-1| + |x+2|$$

* Continuous at $x=1$.

$$f(x) = \begin{cases} -(x-1) - (x+2) = -2x-1 & ; \text{ for } x < -2 \\ -(x-1) + (x+2) = 3 & ; \text{ for } -2 \leq x < 1 \\ (x-1) + (x+2) = 2x+1 & ; \text{ for } x \geq 1 \end{cases}$$

$$\text{At } x=1 ; f(1) = 2(1)+1 = 3$$

$$\text{LHL} = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \{3\} = 3$$

$$\text{RHL} = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \{2x+1\} = 3$$

$$\therefore \text{LHL} = \text{RHL} = f(1) = 3$$

$\therefore f(x)$ is continuous at $x=1$.

* Differentiability at $x=-2$.

$$\begin{aligned} \text{LHD} &= \lim_{h \rightarrow 0^-} \frac{f(-2+h) - f(-2)}{-h} \\ &= \lim_{h \rightarrow 0^-} \frac{(3-2h) - (3)}{-h} \\ &= 2 \end{aligned}$$

$$\text{At } x < -2$$

$$f(x) = -2x-1$$

$$\begin{aligned} f(-2+h) &= -2(-2+h)-1 \\ &= 4-2h-1 \\ &= 3-2h \end{aligned}$$

$$f(-2) = 3.$$

$$\begin{aligned} \text{RHD} &= \lim_{h \rightarrow 0^+} \frac{f(-2+h) - f(-2)}{h} \\ &= \lim_{h \rightarrow 0^+} \left\{ \frac{3-3}{h} \right\} \\ &= 0 \end{aligned}$$

$$\text{At } x > -2$$

$$f(x) = 3$$

$$f(-2+h) = 3$$

$$f(-2) = 3$$

$$\therefore \text{LHD} \neq \text{RHD}$$

$f(x)$ fails to be differentiable at $x=-2$ //

$$34) b) x = \sin t, y = \sin pt ;$$

D w.r.t t

$$\frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = \cos pt \cdot (p) ; \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{p \cdot \cos pt}{\cos t}$$

D w.r.t x on B.S

$$\frac{d^2y}{dx^2} = \frac{\cos t (-\sin pt)(p) + p \cos pt \sin t}{(\cos^2 t)} \cdot \frac{dt}{dx}$$

$$\therefore \frac{dx}{dt} = \cos t$$

$$\frac{dt}{dx} = \frac{1}{\cos t}$$

$$\therefore \frac{dy}{dx} = \left(\frac{p \cos pt}{\cos t} \right)$$

$$= \frac{-p^2 (\sin pt)(\cos t) + p(\cos pt)(\sin t)}{(\cos^3 t)}$$

$$= \frac{-p^2 \sin pt \cos t}{(\cos^3 t)} + \frac{p(\cos pt)(\sin t)}{(\cos^3 t)}$$

$$= \frac{-p^2(y)}{\cos^2 t} + \frac{x \left(\frac{dy}{dx} \right)}{\cos^2 t}$$

$$\frac{d^2y}{dx^2} = \frac{-p^2 y + x \frac{dy}{dx}}{\cos^2 t}$$

$$(\cos^2 t) \cdot \frac{d^2y}{dx^2} = -p^2 y + x \frac{dy}{dx}$$

$$(1 - \sin^2 t) \cdot \frac{d^2y}{dx^2} = -p^2 y + x \frac{dy}{dx}$$

$$(1 - x^2) \cdot \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2 y = 0$$

Here proved //

Q:35 $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$
 let $I = \int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$ — (1)

$\int_0^a f(x) dx = \int_0^a f(a-x)$ property.

$$I = \int_0^{\pi/2} \frac{(\frac{\pi}{2}-x) \sin(\frac{\pi}{2}-x) \cos(\frac{\pi}{2}-x)}{\sin^4(\frac{\pi}{2}-x) + \cos^4(\frac{\pi}{2}-x)} dx$$

$$I = \int_0^{\pi/2} \frac{(\frac{\pi}{2}-x) \cos x \sin x}{\cos^4 x + \sin^4 x} dx \quad \text{--- (2)}$$

① + ② $\Rightarrow$

$$2I = \int_0^{\pi/2} \frac{\sin x \cos x \left[x + \frac{\pi}{2} - x \right]}{\sin^4 x + \cos^4 x} dx$$

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$$

$\div$ Ng, Div of RHS by $\cos^4 x$.

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{\tan x \sec^2 x}{1 + \tan^4 x} dx$$

$$2I = \frac{\pi}{2} \times \frac{1}{2} \int_0^{\pi/2} \frac{2 \tan x \sec^2 x}{1 + \tan^4 x} dx$$

$$2I = \frac{\pi}{4} \int_0^{\pi/2} \frac{2 \tan x \sec^2 x}{1 + (\tan^2 x)^2} dx$$

Put $\tan^2 x = t$

$\Rightarrow$ w.r.t x

$2 \tan x \sec^2 x dx = dt$

when $x=0, t=0$

$x=\pi/2, t=\infty$

$$2I = \frac{\pi}{4} \int_0^{\infty} \frac{1}{1+t^2} dt$$

$$2I = \frac{\pi}{4} \left[\tan^{-1} t \right]_0^{\infty} = \frac{\pi}{4} \{ \tan^{-1} \infty - \tan^{-1} 0 \}$$

$$2I = \frac{\pi}{4} \left\{ \frac{\pi}{2} - 0 \right\} = \frac{\pi^2}{8}$$

$$I = \frac{\pi^2}{16}$$

$$\begin{aligned}
 & 35) b) \int \frac{1}{\sin x + \sin 2x} dx \\
 & \text{(OR)} = \int \frac{1}{\sin x + 2 \sin x \cos x} dx \\
 & = \int \frac{\sin x}{\sin x [\sin x + 2 \sin x \cos x]} dx \\
 & = \int \frac{\sin x}{\sin^2 x [1 + 2 \cos x]} dx \\
 & = \int \frac{-dt}{(1-t^2)(1+2t)} \\
 & = \int \frac{-1 dt}{(1-t)(1+t)(1+2t)} \quad \text{Partial fraction method}
 \end{aligned}$$

$$\begin{aligned}
 & \text{Let } \boxed{\cos x = t} \\
 & \text{D w.r.t } x. \\
 & -\sin x = \frac{dt}{dx} \\
 & \boxed{\sin x dx = -dt}
 \end{aligned}$$

$$\begin{aligned}
 & \because \sin^2 x = 1 - \cos^2 x \\
 & \boxed{\sin^2 x = 1 - t^2}
 \end{aligned}$$

$$\Rightarrow \frac{-1}{(1-t)(1+t)(1+2t)} = \frac{A}{(1+t)} + \frac{B}{(1-t)} + \frac{C}{(1+2t)} \quad \text{--- (1)}$$

$$-1 = A(1-t)(1+2t) + B(1+t)(1+2t) + C(1-t)(1+t)$$

$$0t^2 + 0t - 1 = -1 = (-2A + 2B - C)t^2 + (A + 3B)t + (A + B + C)$$

$$\begin{aligned}
 \text{On equating} \quad & \left. \begin{aligned} -2A + 2B - C &= 0 \\ A + 3B &= 0 \\ A + B + C &= -1 \end{aligned} \right\} \text{--- (2)}
 \end{aligned}$$

$$\begin{aligned}
 & \text{Solving we get} \\
 & \text{(2)} \quad \begin{aligned} A &= \frac{1}{2} \\ B &= -\frac{1}{6} \\ C &= -\frac{4}{3} \end{aligned}
 \end{aligned}$$

$$\therefore \text{Eqn (1)} \Rightarrow \frac{1}{(1+t)(1-t)(1+2t)} = \frac{1}{2} \cdot \frac{1}{(1+t)} - \frac{1}{6} \cdot \frac{1}{(1-t)} + \frac{-4}{3} \cdot \frac{1}{(1+2t)}$$

$$\begin{aligned}
 \int \frac{-1}{(1+t)(1-t)(1+2t)} dt &= \frac{1}{2} \int \frac{1}{1+t} dt - \frac{1}{6} \int \frac{1}{1-t} dt - \frac{4}{3} \int \frac{1}{1+2t} dt \\
 &= \frac{1}{2} \log|1+t| - \frac{1}{6} \log|1-t| - \frac{4}{3} \times \frac{2}{2} \int \frac{1}{1+2t} dt \\
 &= \frac{1}{2} \log|1+t| - \frac{1}{6} \log|1-t| - \frac{2}{3} \int \log|1+2t|
 \end{aligned}$$

$$\text{where } t = \cos x$$

$$36) i) a) 2^{n(B) \times n(G)} = 2^{3 \times 2} = 2^6 \leftarrow \text{Relations from } B \text{ to } G$$

$$b) \text{ Relations from } G \text{ to } B = 2^{n(G) \times n(B)} = 2^{2 \times 3} = 2^6$$

$$ii) \text{ Functions from } B \text{ to } G = n(G)^{n(B)} = 2^3 = 8$$

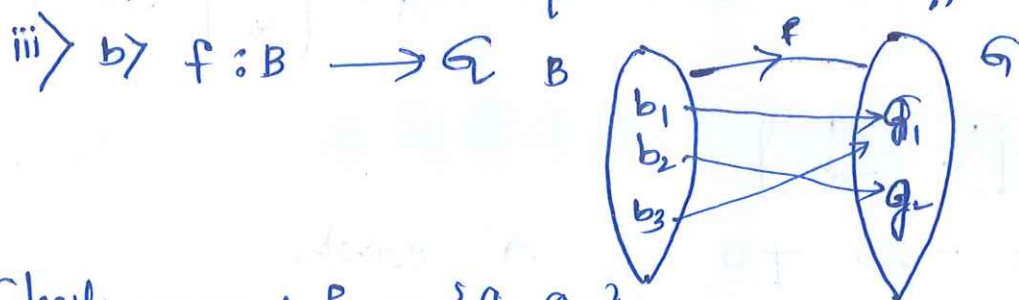
iii) a) Let $R: B \rightarrow B$, $R = \{(x, y) : x \text{ \& } y \text{ are students of the same sex}\}$

* Reflexive : $(b_1, b_1) \in R$

* Symmetric : If $(b_1, b_2) \in R$ then $(b_2, b_1) \in R$.

* Transitive : If $(b_1, b_2) \in R$ and $(b_2, b_3) \in R$ then $(b_1, b_3) \in R$

$\therefore R$ is an equivalence relation //



Clearly range of $f = \{g_1, g_2\}$

Codomain of $f = \{g_1, g_2\}$

$\therefore f$ is ONTO.

* f is NOT one-one :

because element ' g_1 ' has two pre-images (b_1 & b_3)

* f is not one-one

f is onto

* f is not bijective //

$$37) i) a) (x-8)(y+10) = xy$$

$$xy + 10x - 8y - 80 = xy$$

$$10x - 8y = 80$$

$$5x - 4y = 40 \quad \text{--- (i)}$$

$$(x+16)(y-10) = xy$$

$$xy - 10x + 16y - 160 = xy$$

$$10x - 16y = -160$$

$$5x - 8y = -80 \quad \text{--- (ii)}$$

$$b) \text{ Linear eqn : } 5x - 4y = 40$$

$$5x - 8y = -80$$

$$AX = B \quad ; \quad A = \begin{bmatrix} 5 & -4 \\ 5 & -8 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} ; B = \begin{bmatrix} 40 \\ -80 \end{bmatrix}$$

$$ii) A = \begin{bmatrix} 5 & -4 \\ 5 & -8 \end{bmatrix}$$

$$|A| = -20 \neq 0 \quad , \quad A^{-1} \text{ exists}$$

$$\text{adj } A = \begin{bmatrix} -8 & -5 \\ 4 & 5 \end{bmatrix}^T = \begin{bmatrix} -8 & 4 \\ -5 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot \text{adj } A = \frac{1}{-20} \begin{bmatrix} -8 & 4 \\ -5 & 5 \end{bmatrix}$$

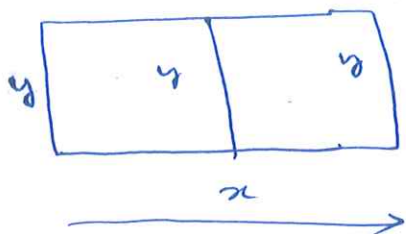
$$X = A^{-1} \cdot B = \frac{1}{-20} \begin{bmatrix} -8 & 4 \\ -5 & 5 \end{bmatrix} \begin{bmatrix} 40 \\ -80 \end{bmatrix} = \frac{-1}{20} \begin{bmatrix} -640 \\ -600 \end{bmatrix}$$

$$X = \begin{bmatrix} 32 \\ 30 \end{bmatrix}$$

$$x = 32, \quad y = 30$$

$$// \quad \text{Amount distributed} \\ = \underline{\underline{2600}}$$

38)



i) Total boundary length $\Rightarrow 2x + 3y = 300$

ii) Area $= l \times b = xy = x \left\{ \frac{300 - 2x}{3} \right\}$
 $= x \left\{ 100 - \frac{2}{3}x \right\}$

iii) a) Area $= f(x) = x \left\{ 100 - \frac{2}{3}x \right\}$

$$f'(x) = 100 - \frac{2}{3}(2x)$$

$$f'(x) = 0 \Rightarrow 100 - \frac{4x}{3} = 0$$

$$300 - 4x = 0$$

$$4x = 300$$

$$x = 75$$

$$y = 50$$

$$\text{Area} = xy = 75 \times 50 = 3750 \text{ sq unit}$$

(OR)

iii) b) As $f'(x) = 100 - \frac{4}{3}x$

$$\text{for } x < 75 \Rightarrow f'(x) > 0$$

$$\text{for } x > 75 \Rightarrow f'(x) < 0$$

function changes from increasing to decreasing
at $x = 75$.

$\therefore$ Max area is at $x = 75$

Hence max area is 3750 sq units //

seen
✓

